

CSL Case Competition

Team UMN

Parul Chaudhary, Bhavisha Chafekar

Saloni Jain, Yehan Wang

Trusted AI Coach for First-Time Homebuyers

Empowering First-Time Homebuyers with Transparent, Explainable AI for PNC Bank Customers



Navigating First-Time Homebuyer Challenges

Key obstacles create confusion and anxiety during the savings and buying process



Overwhelming information on savings options and programs



Uncertainty from shifting home prices and mortgage rates



Hidden costs beyond down payments increase financial pressure



Emotional stress and unclear program details add confusion



Concerns about risk tolerance for 2–3 year savings timeline



Strong desire for transparent, unbiased guidance and support



Understanding Customer Personas & Building AI Trust

Key user profiles and trust factors guiding AI design for better engagement

Customer Personas

Aspirational Millennials (28–35, \$65K–\$95K income): Digitally savvy but overwhelmed by conflicting advice

Entrepreneurial Couples (30–38, \$80K–\$150K income): Require strategic plans to manage irregular earnings

AI Trust Factors

Transparent, clear recommendations

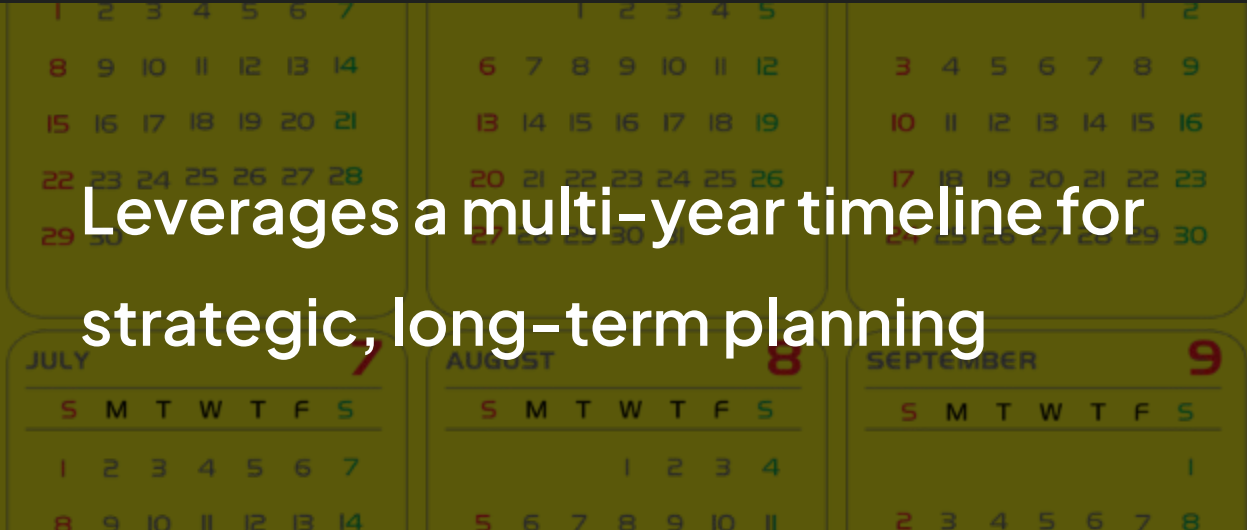
Avoidance of bias in outputs

Visibility into personalized input data

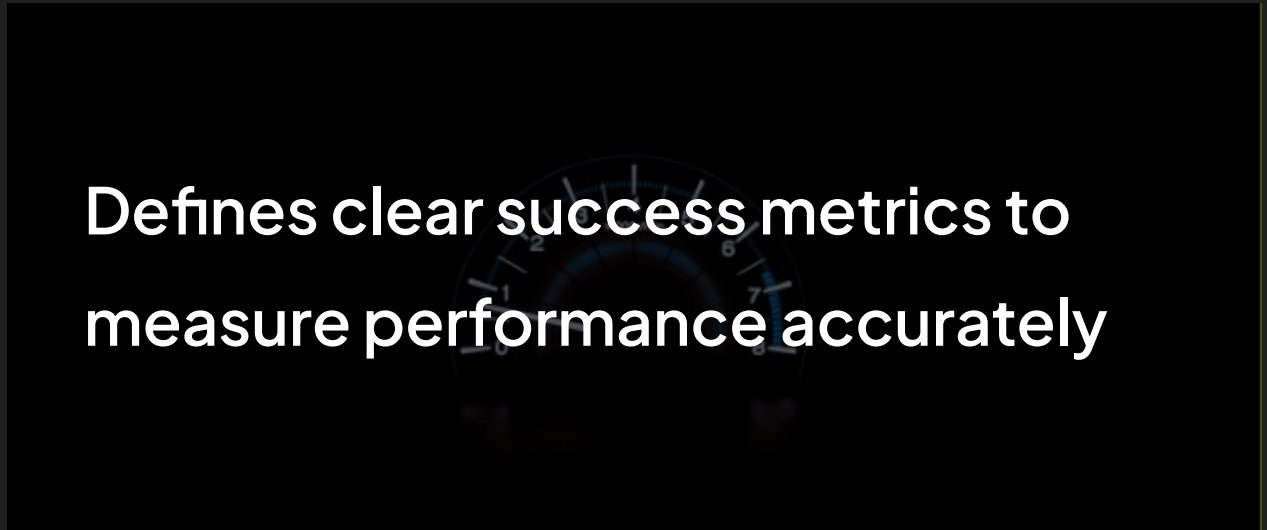
Flexible user overrides for control

Unique Advantages and Trust Signals in AI

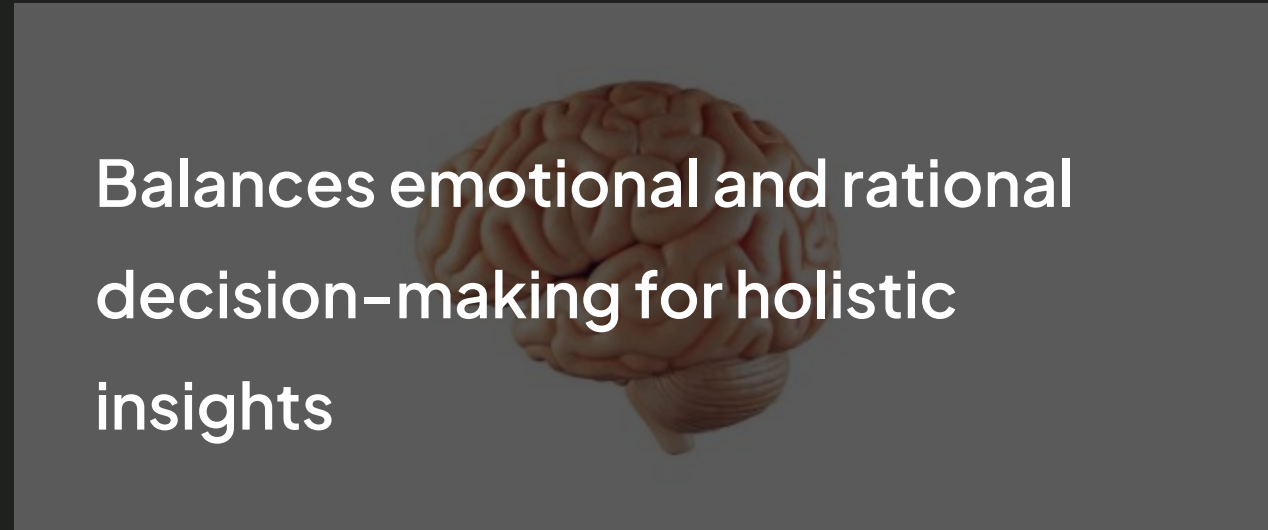
Multi-year timelines and transparent metrics build lasting customer confidence



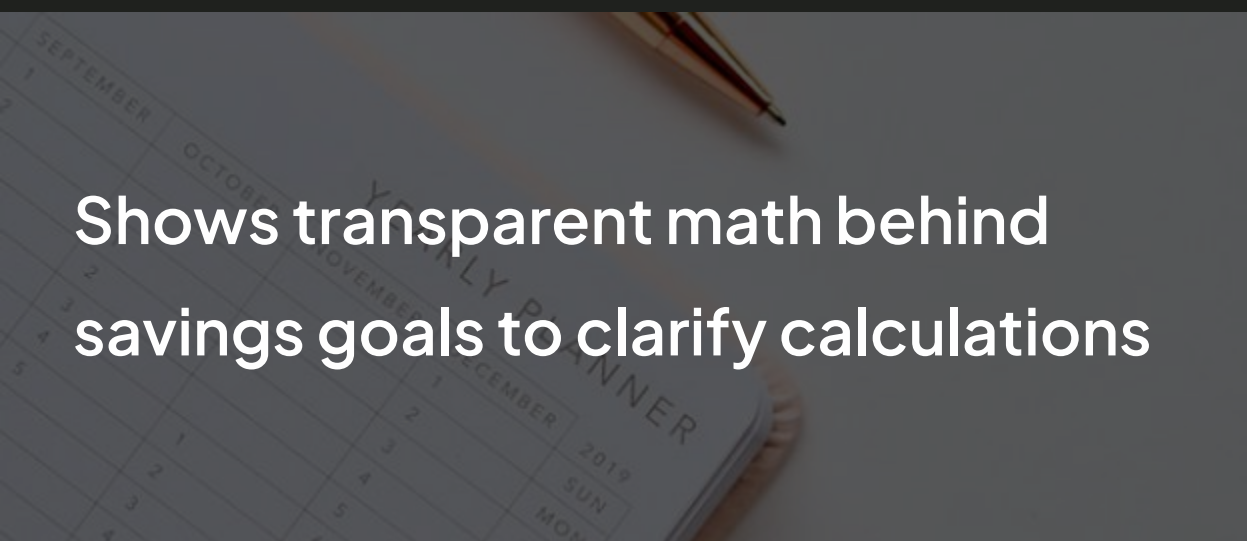
Leverages a multi-year timeline for strategic, long-term planning



Defines clear success metrics to measure performance accurately



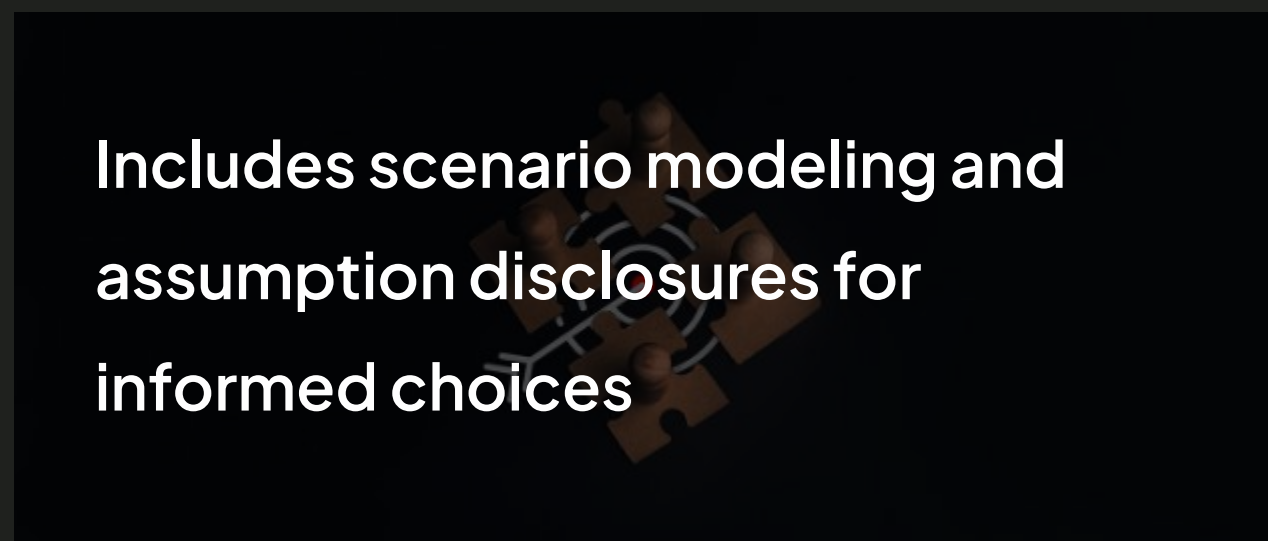
Balances emotional and rational decision-making for holistic insights



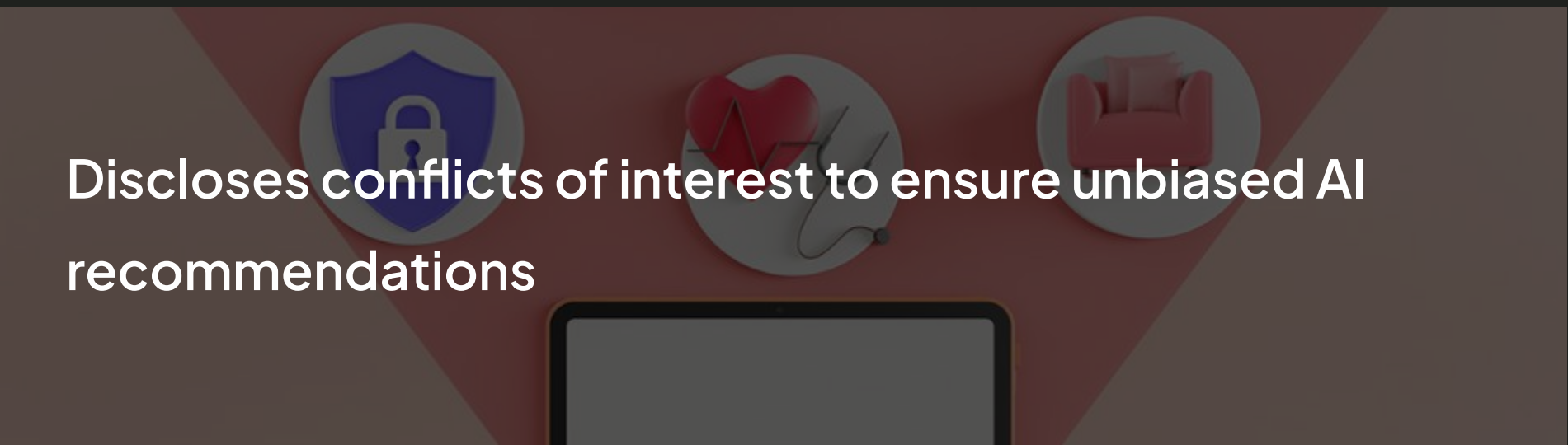
Shows transparent math behind savings goals to clarify calculations



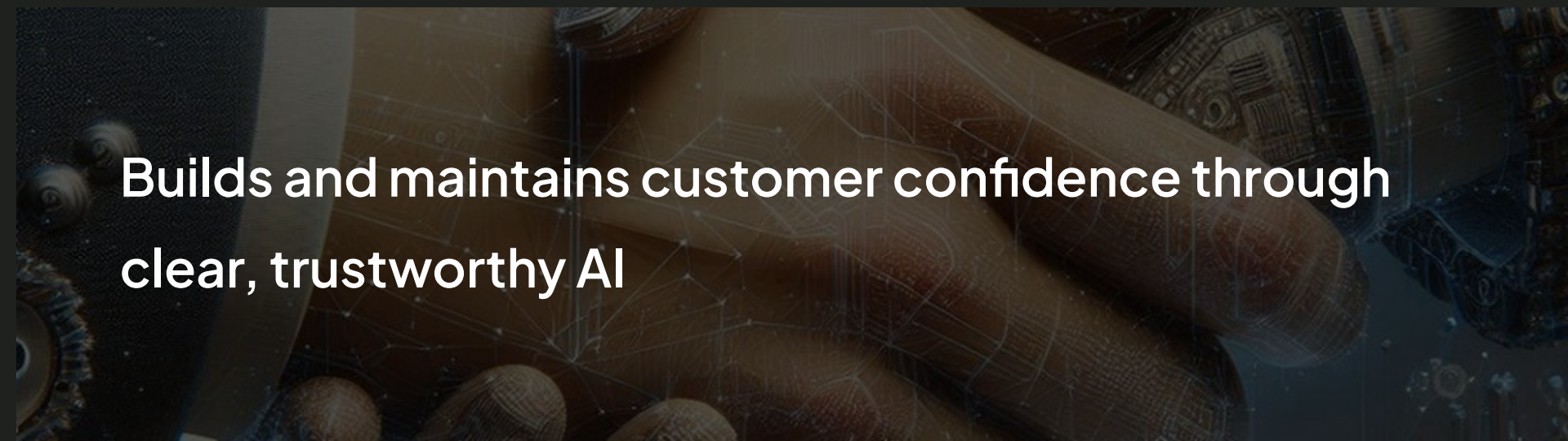
Provides confidence levels on market projections to indicate reliability



Includes scenario modeling and assumption disclosures for informed choices



Discloses conflicts of interest to ensure unbiased AI recommendations



Builds and maintains customer confidence through clear, trustworthy AI

Navigating Early Customer Phases: From Aspiration to Goal Setting

How AI Resolves Confusion and Guides Financial Decisions in Home Buying

Phase 1: Aspiration & Confusion (Months 1–3)

Buyers feel pressure from peers buying, face conflicting information, and lack tailored local guidance. AI clarifies realistic price ranges and transparently explains pricing calculations.



Phase 2: Goal Setting & Reality Check (Months 3–6)

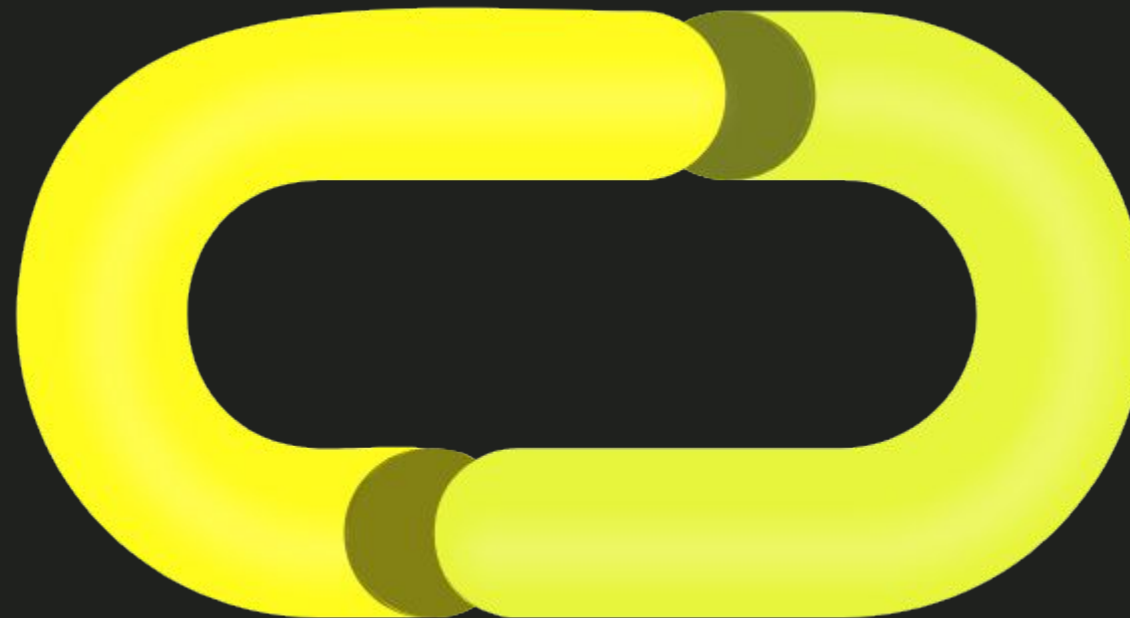
Confusion over down payments, PMI, and balancing savings versus investments is common. AI models trade-offs among three down payment options, helping customers choose balanced plans with clear financial impacts.

Customer Journey: Phases 3 & 4 — Savings & Market Timing

AI-driven support guides buyers through saving execution and market timing decisions

Phase 3: Savings Execution & Optimization (Months 6–18)

Buyers track progress, face unexpected expenses, and reconsider strategies. AI offers monthly check-ins, suggests improved savings options, alerts budget variances, and allows overrides while disclosing product biases.



Phase 4: Market Monitoring & Timing Anxiety (Months 18–24)

Approaching savings goals, buyers worry about price jumps and rate spikes. AI evaluates readiness, compares buy-now versus wait scenarios using local market data, and recommends timing based on financial and psychological factors with transparent confidence levels.



Empowering Confident Pre-Purchase Decisions

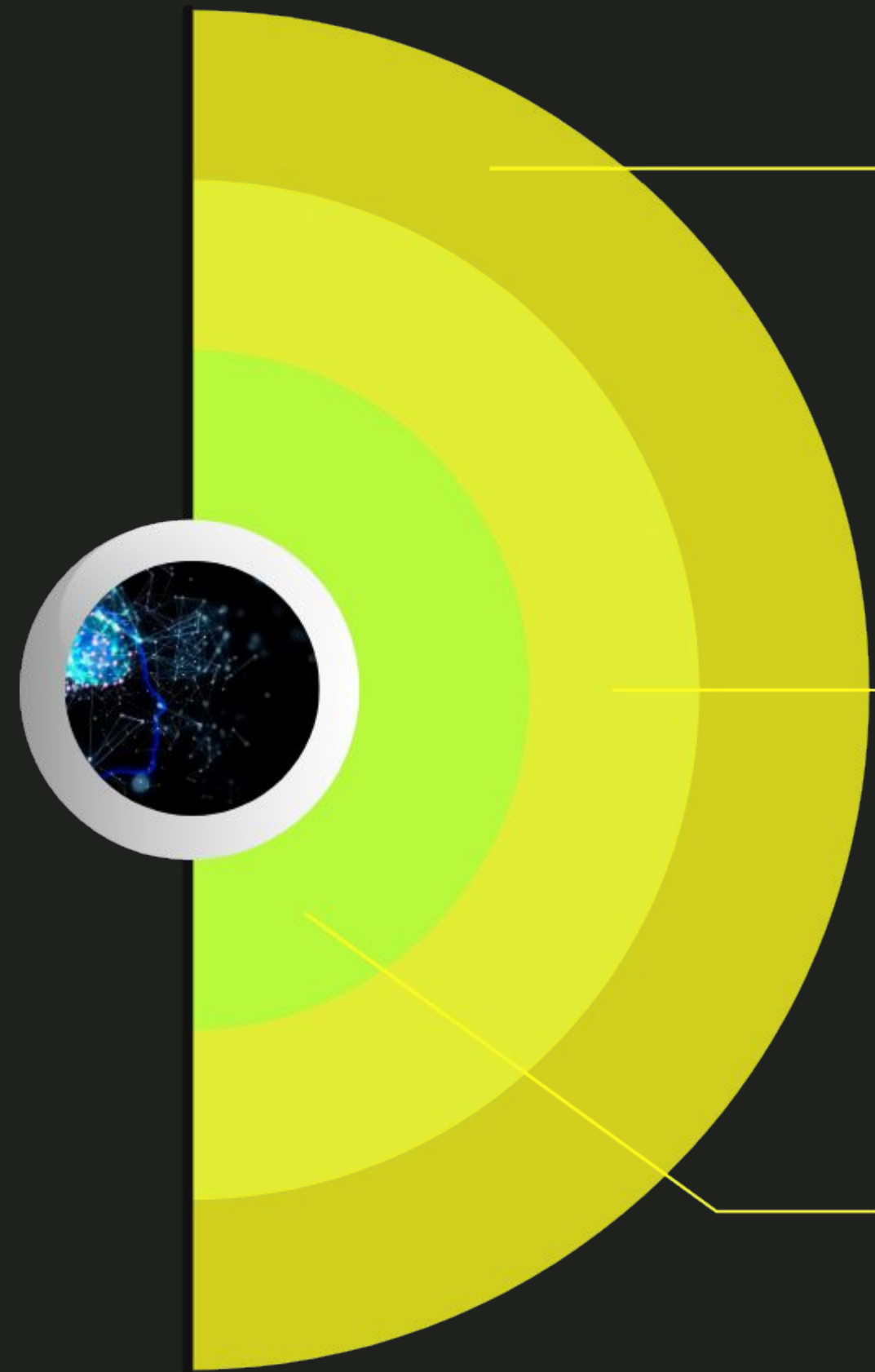
AI-Driven Insights on Offers, Market Values, and Risk Transparency

Offer Scenario	Listing Price Evaluation	Inspection Repair Impact	Closing Costs (\$)	Acceptance Probability (%)	Long-Term Value Projection	Risk Transparency
Single Offer	Market average	Minimal repairs needed	4,500	75	Stable growth	Low risk, full disclosure
Multiple Offers	Above market	Moderate repairs required	5,200	60	High potential value	Medium risk, detailed alerts
Negotiated Offer	Below market	Significant repairs expected	3,800	85	Moderate growth	Higher risk, override options

Clear UX Flow & Bias

Transparency

Empowering users with clarity,
fairness, and control through layered
design and bias monitoring



Progressive Disclosure

Delivers simple progress updates, key factor details on demand, and full transparency of AI calculations.

Visible Bias Dashboard

Monitors fairness continuously with frequent audits and real-time alerts to maintain ethical AI use.

User Override with Explanation

Allows users to override AI recommendations with supported explanations, promoting trust and empowerment.

Securing Growth: Governance & Phased Market Launch

Implementing robust data controls and a strategic rollout to ensure privacy, compliance, and market success

Data Privacy & Controls

Ensures user privacy with granular access controls and continuous audit logging to protect sensitive information.

Risk Mitigation

Addresses technical, regulatory, and reputational risks to ensure smooth deployment and sustained market trust.



Bias Testing & Model Oversight

Applies multi-layer bias testing and manages model lifecycle to maintain fairness and reliability in AI outputs.

Phased Market Launch

Rolls out in stages: pilot in Q4 2025 targeting 500 customers, limited launch in Q1 2026 across 5 cities with enhanced features, followed by national rollout from Q3 2026 integrating AI in mobile apps and partnerships.

Driving Value Through Clear Benefits and Measurable Success

Aligning Customer, PNC, and Societal Gains with Impactful Metrics

Customers	PNC Bank	Society & Success Metrics
Clear guidance and unbiased advocacy	Increased mortgage revenue and \$50M new originations	Improved financial inclusion, literacy, and consumer protection
Increased confidence and control	Risk reduction and regulatory goodwill	Regulatory leadership and zero bias incidents
Continuous financial learning	Differentiation and higher customer lifetime value	40% on-time savings goal achievement
		25% mortgage pre-approval rate